

STADIOEquities — Data Request

Project Early Account Activation Prediction
Module CAP182 Capstone Project SS1
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Purpose of this request

To create a model to predict if a newly registered account will make its first deposit by 30 days after registration, we need historical data as a part of the account (not per user), such as how many logins each account has, or how many transactions each account has conducted; who interacted with it, how it went from the point of being picked up, what it did during its first session, and whether and when it was first funded.

Storing grain and its past

- For the account-level tables (Tables 1, 3, 4), only one row per registered account.
- For the behavioural table (Table 2), keyed to `account_id`, one row per event is required. The following features will be abstracted from the first session.
- Please provide at least the past 4 years of registrations, which corresponds to the 4 years of the app/web.
- Negative cases are absolutely necessary, as they are found in behaviour history, including accounts that *never* funded.
- Each table must have a single, stable `account_id` that is used as the join key, resolving the app and web sessions per account.

Data handling / privacy (POPIA)

- Please include data pseudonymised — no names, no ID numbers, no email addresses, no phone numbers or free text that could identify an individual. The `account_id` should be a surrogate key, not a real client identifier.
- Region at province level is enough; no street-level location is needed.
- Data can be provided in either CSV or Parquet format. For the behavioural event table the preferred format is Parquet, given its volume.

Table 1. Account registration & funding (one row per account; includes the target)

Column name	Data type	Example	Additional information
Account ID	String (surrogate key)	ACC0000481923	Unique per account. Single key shared by all tables. Must not be an actual client identifier.

Column name	Data type	Example	Additional information
Registration timestamp	String, datetime	2024-03-11 08:42:19	Format <code>yyyy-MM-dd HH:mm:ss</code> . Indicates the beginning of the 30-day activation period.
Registration platform	String, categorical	app	Valid values: <code>app</code> , <code>web</code> .
KYC status	String, categorical	verified	Indicates whether KYC is complete or ongoing. Valid values: not started, in progress, verified, abandoned, rejected.
KYC completion timestamp	String, datetime	2024-03-11 09:15:02	Format <code>yyyy-MM-dd HH:mm:ss</code> . Null if KYC does not complete.
Onboarding steps completed	Numeric, integer	5	Number of steps completed in onboarding (0 – total steps). Please also provide the number of steps altogether.
Steps in onboarding flow (total)	Numeric, integer	7	Total number of steps in the onboarding flow entered at that time.
Onboarding abandoned flag	Boolean	false	<code>true</code> if the client abandoned onboarding before completing it.
First deposit timestamp	String, datetime	2024-03-14 19:03:55	Format <code>yyyy-MM-dd HH:mm:ss</code> . Null if the account made no deposits.
First deposit amount	Numeric, integer	5000	Amount in cents (e.g. <code>5000</code> = R50.00). Null if there was never a deposit.
Qualifying deposit flag	Boolean	true	<code>true</code> if the deposit was successful, not reversed, and constituted a qualifying deposit. Clarifies Issue I1.
Activated within 30 days	Boolean	true	Target variable. <code>true</code> if a qualifying first deposit was made within 30 days after registration, else <code>false</code> . If you cannot calculate this, provide the timestamps above and we will calculate it.
Account status (current)	String, categorical	funded_active	Valid values: <code>registered_unfunded</code> , <code>funded_active</code> , <code>dormant</code> , <code>closed</code> . For validation purposes only; not a model feature.

Table 2. Onboarding & first-session behaviour (one row per event, keyed to account_id)

Column name	Data type	Example	Additional information
Site ID	UUID	123e47e8-a1df-4b2e-9cde-bd426442785f	Unique identifier for the site.
Account ID	String	ACC0000481923	Foreign key to Table 1.
Session ID	String	SESS_9f21ab	Groups events together in the same session.

Column name	Data type	Example	Additional information
Event timestamp	String, datetime	2024-03-11 08:44:07	Format yyyy-MM-dd HH:mm:ss . The first 3 days after registration are strictly required (to avoid look-ahead); more is welcome.
Event type	String, categorical	screen_view	Any valid string, such as screen_view , feature_use , onboarding_step , deposit_attempt , search , error .
Screen / feature name	String, categorical	deposit_screen	Name of the screen or feature in which the event occurred.
Session platform	String, categorical	app	Valid values: app , web . Facilitates solving web/app identity fragmentation.
Session sequence number	Numeric, integer	1	1 = the client's first ever session, 2 = second, etc.
Event duration (seconds)	Numeric, integer	12	Time spent on the screen/feature, if applicable. Null if not tracked.

Table 3. Basic client information (one row per client; captured at sign-up)

Column name	Data type	Example	Additional information
Account ID	String	ACC0000481923	Foreign key to Table 1.
Age at registration	Numeric, integer	29	In years. Valid range roughly 18 – 100. Care must be taken with its use in relation to POPIA / fairness.
Province	String, categorical	GP	Nine provinces: GP , EC , FS , KZN , LP , MP , NC , NW , WC .
Stated investment goal	String, categorical	long_term_growth	Valid values e.g. long_term_growth , save_regularly , short_term_trading , learning , unspecified .
Stated risk appetite	String, categorical	moderate	Valid values: low , moderate , high , unspecified .
Preferred language	String, categorical	en	The user's preferred language as an ISO-style code (e.g. en , af , zu , xh). Optional.

Table 4. Marketing acquisition (one row per account)

Column name	Data type	Example	Additional information
Account ID	String	ACC0000481923	Foreign key to Table 1.
Acquisition channel	String, categorical	paid_social	Valid values e.g. paid_social , paid_search , organic , referral , influencer , email , other .
Campaign ID	String	CMP_2024Q1_042	Identifier of the campaign that acquired the account. Null for organic.

Column name	Data type	Example	Additional information
Acquisition cost	Numeric, integer	18000	Cost to acquire this account in cents (e.g. 18000 = R180.00).
Referral flag	Boolean	false	true if the account was acquired via an existing client referral.
First 30-day nudges sent	Numeric, integer	4	Count of onboarding emails/nudges sent in the first 30 days. Context for the current fixed-schedule approach.
First 30-day nudge opens	Numeric, integer	1	Count of those nudges opened / engaged with.

Notes on data type choices

Data type choices follow the conventions in the IDS182 and WWD182 study guides: identifiers and categoricals as **strings**, dates/times as **datetime strings** in yyyy-MM-dd HH:mm:ss format, counts and monetary amounts as **integers** (money in cents to avoid floating-point rounding), and yes/no fields as **booleans**. Where a column has a fixed set of valid values, these are listed in the *Additional information* column so the data can be validated on delivery.